

Decisions Log

This document records key decisions made during the EMS Readiness Optimization project.

Decision Template

[DEC-XXX] Decision Title

Date: YYYY-MM-DD

Decision Maker: [Name/Role]

Status: Proposed / Accepted / Superseded

Context:
[What situation prompted this decision?]

Options Considered:

- Option A: [Description]
- Option B: [Description]
- Option C: [Description]

Decision:
[Which option was chosen and why]

Consequences:
[What are the implications of this decision?]

Decisions

[DEC-001] Study Area Selection

Date: 2026-03-12

Decision Maker: Project Lead

Status: Accepted

Context:

Need to select primary and secondary study areas for the EMS optimization analysis.

Options Considered:

- All of NYC: Comprehensive but computationally expensive
- Manhattan only: Focused, manageable scope
- Manhattan + CBD robustness check: Balanced approach

Decision:

Option 3 - Manhattan as primary study area with CBD (MTA Congestion Relief Zone) as secondary area for robustness testing.

Consequences:

- Allows focused analysis with reasonable computational requirements
- CBD provides natural robustness check for high-demand area
- Results may not generalize to outer boroughs

[DEC-002] Staging Location Candidates

Date: 2026-03-12

Decision Maker: Project Lead

Status: Accepted

Context:

Need to define candidate locations for EMS staging.

Options Considered:

1. Grid-based locations: Uniform coverage
2. Hospital locations: Practical operational sites
3. FDNY firehouses: Existing infrastructure
4. Combined approach: Multiple location types

Decision:

Option 3 - Use FDNY firehouses as staging candidates.

Consequences:

- Leverages existing infrastructure
- Realistic operational scenario
- Limited to current firehouse locations (may not be optimal in absolute terms)

[DEC-003] Simulation Framework

Date: 2026-03-12

Decision Maker: Project Lead

Status: Accepted

Context:

Need to select simulation methodology for EMS system modeling.

Options Considered:

1. Agent-based simulation: Detailed but complex
2. Discrete-event simulation (DES): Standard for queueing systems
3. System dynamics: Too aggregated for this application

Decision:

Option 2 - Discrete-event simulation using SimPy.

Consequences:

- Well-suited for modeling arrival, dispatch, service processes
- Efficient computation
- Python ecosystem enables integration with optimization

[DEC-004] Optimization Approach

Date: 2026-03-12

Decision Maker: Project Lead

Status: Accepted

Context:

Need to select optimization method for ambulance staging.

Options Considered:

1. Heuristic approaches: Fast but no optimality guarantee
2. Mixed-integer programming: Optimal but potentially slow
3. Linear programming (relaxed): Fast, near-optimal for many problems
4. Metaheuristics: Flexible but tuning required

Decision:

Option 3 - Linear programming using PuLP, with integer constraints if needed.

Consequences:

- Fast solution times
 - Optimality guarantees (for LP formulation)
 - May need to round/adjust for integer ambulance counts
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[DEC-005] Data Temporal Scope

Date: 2026-03-12

Decision Maker: Project Lead

Status: Accepted

Context:

Need to determine which years of crash data to use.

Options Considered:

1. Most recent year only: Current patterns
2. Last 3 years: Balances recency and stability
3. All available data: Maximum sample size

Decision:

Option 3 - Use all available data for demand pattern estimation, with options to filter by date range.

Consequences:

- Maximum statistical power for pattern estimation
 - May include outdated patterns (e.g., pre/post COVID)
 - Can subset for sensitivity analysis
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[DEC-008] Alternative Distance Metric — Manhattan (Taxicab) Distance

Date: 2026-03-12

Decision Maker: Technical Lead

Status: Accepted

Context:

Haversine (great-circle) distance underestimates true road distances in a grid-based street network

like Manhattan. Need to evaluate whether a Manhattan (taxicab) distance metric produces materially different optimization and simulation results.

Options Considered:

1. Haversine only (status quo)
2. Manhattan (taxicab) distance replacing Haversine
3. Both metrics compared side-by-side with simulation evaluation
4. Road network distance (OSRM / Google routing)

Decision:

Option 3 — Implement Manhattan distance as an alternative metric, run P2 optimization with both, and compare via simulation.

Rationale:

- Manhattan distance is ~27% longer on average than Haversine for Manhattan geography, better approximating grid-based street travel
- Side-by-side comparison quantifies sensitivity to metric choice
- Road network routing (Option 4) out of scope — requires external APIs and adds complexity without changing relative policy rankings
- Simulation dispatcher always uses Haversine internally, so this isolates the effect on allocation decisions

Consequences:

- Added `manhattan_distance()` function to `distance.py` and `metric` parameter to `build_distance_matrix()`
- Manhattan matrix generated: mean 5.48 mi vs. Haversine 4.29 mi
- Experiment showed near-identical simulation results (mean RT 2.55 min for both), confirming P2 allocation is robust to distance metric choice
- Only 2 of 48 firehouses differ between allocations
- Validates that Haversine is a sufficient approximation for this study

[DEC-009] CBD-Focused vs. Manhattan-Wide Optimization

Date: 2026-03-12

Decision Maker: Technical Lead

Status: Accepted

Context:

The CBD (precincts 1, 5, 6, 7, 9, 10, 13, 14, 17, 18) accounts for ~56% of total demand. Need to evaluate whether a CBD-focused optimization strategy (up-weighting CBD demand) improves CBD response times and is preferable to the Manhattan-wide P2 model.

Options Considered:

1. Manhattan-wide P2 only (status quo)
2. CBD-only model (ignore non-CBD precincts)
3. CBD-focused model with 3× demand weight on CBD precincts
4. Multi-objective (Pareto) CBD/non-CBD optimization

Decision:

Option 3 — Implement CBD-focused demand-weighted model (3× CBD weight) and compare against Manhattan-wide P2 via simulation.

Rationale:

- CBD-focused (3× weight) tests whether concentrating resources improves CBD performance
- Comparison with Manhattan-wide P2 quantifies equity trade-offs
- CBD-only model (Option 2) is impractical — abandons non-CBD coverage
- Multi-objective Pareto (Option 4) deferred to future work

Consequences:

- Added `build_cbd_focused_demand_weighted()` and `build_cbd_focused_coverage()` to `models.py`
- CBD-focused P2 does NOT improve CBD RT (2.50 vs. 2.47 min, +1.2%) but dramatically worsens non-CBD RT (6.88 vs. 2.66 min, +159%)
- Non-CBD 8-min coverage drops from 100% to 73.6%
- Validates that Manhattan-wide P2 already near-optimally serves the CBD without sacrificing equity
- Recommends against CBD-focused optimization for operational deployment

Last Updated: March 12, 2026

[DEC-003] Optimization Policy Selection

Date: 2026-03-12

Decision Maker: Technical Lead

Status: Accepted

Context:

Multiple optimization formulations exist for EMS allocation. Need to determine which models to implement and compare for Manhattan EMS readiness.

Options Considered:

1. Demand-Weighted Optimization: Minimize demand-weighted average response time
2. P-Median: Minimize total response time (unweighted)
3. Maximal Coverage: Maximize demand covered within threshold
4. Set Covering: Cover all demand with minimum units
5. Hybrid: Multi-objective optimization

Decision:

Implement and compare options 1-3 (Demand-Weighted, P-Median, Maximal Coverage) plus two baseline policies (Uniform, Demand-Proportional).

Rationale:

- Demand-Weighted (P2) aligns with equity goals (weight by demand intensity)
- P-Median (P2b) provides comparison for unweighted optimization
- Maximal Coverage (P2c) represents pure coverage objective
- Baselines (P0, P1) provide non-optimized benchmarks
- Set Covering is subsumed by other models
- Multi-objective deferred to future work

Consequences:

- Need to implement 3 MIP solvers + 2 baseline policies
 - Compare 5 policies × 4 K values = 20 scenarios
 - Provides comprehensive policy comparison for decision-makers
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[DEC-004] Unit Count Scenarios (K)

Date: 2026-03-12

Decision Maker: Technical Lead

Status: Accepted

Context:

Need to determine which unit counts to evaluate for optimization and sensitivity analysis.

Options Considered:

1. $K = \{20, 30, 40, 48\}$ - Current recommendation
2. $K = \{10, 20, 30, 40, 50\}$ - Wider range
3. $K = \{25, 30, 35, 40, 45, 50\}$ - Finer granularity
4. $K = 40$ only - Current Manhattan capacity

Decision:

Use $K = \{20, 30, 40, 48\}$

Rationale:

- $K=20$: Resource-constrained scenario (50% of current)
- $K=30$: Moderate resource scenario (75% of current)
- $K=40$: Current Manhattan capacity (baseline)
- $K=48$: All firehouses staffed with 1 unit (upper bound given capacity=5)
- Provides sufficient range to observe diminishing returns
- Computationally efficient (4 scenarios $\times$ 5 policies = 20 optimizations)

Consequences:

- May miss optimal K if it falls outside range (e.g., $K=25$)
- But results show $K=20$ achieves near-optimal for P2, so range is sufficient
- Future work can refine with continuous K if needed

[DEC-005] Firehouse Capacity Constraint

Date: 2026-03-12

Decision Maker: Technical Lead

Status: Accepted

Context:

Need to determine maximum units that can be allocated to a single firehouse.

Options Considered:

1. No limit (unconstrained)
2. Capacity = 3 units per firehouse
3. Capacity = 5 units per firehouse
4. Capacity = 10 units per firehouse
5. Variable capacity by firehouse size

Decision:

Use capacity = 5 units per firehouse (uniform across all locations)

Rationale:

- Based on FDNY operational guidance (typical firehouse can support 4-6 units)
- Prevents unrealistic concentration (e.g., all 40 units at one location)

- 5 units allows sufficient flexibility for optimized allocation
- Uniform capacity simplifies optimization formulation
- Conservative estimate (actual capacity varies 3-10 by location)

Consequences:

- P2 allocates up to 5 units at high-demand firehouses (Midtown, Financial District)
 - May under-utilize capacity at larger firehouses
 - Future work can incorporate actual capacity data per location
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[DEC-006] Coverage Threshold for Maximal Coverage Model

Date: 2026-03-12

Decision Maker: Technical Lead

Status: Accepted

Context:

Maximal Coverage model requires threshold τ (minutes) to define “covered” demand.

Options Considered:

1. $\tau = 5$ minutes (aggressive response target)
2. $\tau = 8$ minutes (NFPA recommendation)
3. $\tau = 10$ minutes (relaxed target)
4. $\tau = 6$ minutes (NYC local law)
5. Multiple thresholds for sensitivity analysis

Decision:

Use $\tau = 8$ minutes (NFPA standard)

Rationale:

- NFPA 1710 specifies 8 minutes for 90% of emergencies
- Aligns with national standards for comparability
- NYC local law (6 min) is aspirational but not consistently achievable
- All policies achieve 100% coverage at $\tau=8$ with $K \geq 20$
- Using $\tau=5$ would require more units; $\tau=10$ is too lenient

Consequences:

- P2c (Maximal Coverage) achieves 100% coverage but with worse response time (3.48 min)
 - May want to test $\tau=6$ in future work to align with NYC requirements
 - 8-minute threshold appropriate for Manhattan context (dense, short distances)
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[DEC-007] Recommended Policy for Deployment

Date: 2026-03-12

Decision Maker: Technical Lead

Status: Proposed

Context:

After comparing 5 policies, need to recommend best allocation strategy for Manhattan EMS.

Options Considered:

1. P0 (Uniform): Equal distribution

2. P1 (Demand-Proportional): Simple heuristic
3. P2 (Demand-Weighted): Minimizes weighted response time
4. P2b (P-Median): Minimizes total response time
5. P2c (Maximal Coverage): Maximizes coverage

Decision:

Primary recommendation: P2 (Demand-Weighted) with K=20-30

Rationale:

- Best response time: 2.49-2.54 min (0-2% from optimal)
- 100% coverage at $K \geq 20$
- Efficient: Uses only 20-23 firehouses vs. 34+ for other policies
- Significant improvement over P0: 17-86% faster response time
- Fast solve time: <0.05 seconds
- Diminishing returns beyond K=30 (additional units provide <0.01 min benefit)

Alternative: P2b for political feasibility (distributes units more evenly)

Consequences:

- Requires MIP solver integration (PuLP + CBC)
- Concentrated allocation may face political resistance (some firehouses get 0 units)
- Recommend phased implementation: Start with P1, transition to P2
- Quarterly reallocation based on updated demand patterns